1 Elicitation versus teaching

Fine-tuning a model on data D can do two different things. It can *elicit*: the information D carries is already in the model, and training only brings it out. Or it can *teach*: the information is not there and the data must supply it. The two look the same from outside, since in both cases the behaviour appears after training, but they mean different things for safety: an unlearned capability that is elicited by a few facts was never removed, and a trait that transfers through unrelated data must have been held already. This paper asks whether the difference can be read off the model and the data *before* training, and gives scores that place a (model, data) pair on the elicit–teach axis.

Ground truth from the training curve. Donoway et al. [4] measure the difference after the fact, with the elicitation description length (EDL): the code length of the data under prequential fine-tuning, swept over the training-set size n. Teaching shows a rise-then-fall hump in EDL per example (the cost of building the capability), elicitation a curve that falls from the first examples. We sharpen this into a per-run read by repeating each sweep on the same data with the responses shifted across prompts, which carries the format and the training dynamics but no pairing information, and subtracting it: the hump of a taught capability survives the subtraction, a cost paid only for the format does not (a *gate*, crossed after a fixed number of examples), and an elicited capability leaves a curve at or below zero.

A toy task with known answers. We build the ground truth on arithmetic (addition and subtraction of integers in a bare natural-language format: the question, a newline, the answer). The training data, the protocol (LoRA, 19 sizes from 100 to 10^6 examples) and the held-out block are the same for five parents, which differ only in what they already hold (Table 1). A TinyStories-1B model we pretrained on children's stories has never seen arithmetic: training on D must teach it, with or without a format installer that fixes the answer format on random labels and teaches no arithmetic (gated at $\leq 2\%$ exact match after installation). The same model taught arithmetic in an operator notation and then bridged to natural language on already-seen rows (no natural-language answer is ever trained, yet a third of the bare questions are answered correctly through the bridge) holds the operations: training on D elicits them. Llama-3.2-1B answers none of the bare questions but a third of the same questions in a scaffolded format; a bare-format installer (trained on 16 multiplication examples) makes it answer in the bare format (97%) and keeps that accuracy: both elicit. The EDL reads agree with the construction: both TinyStories parents TEACH (hump peak near $n = 215{,}000$), the bridge parent and Llama base are GATE (a format cost paid at the start), the installed Llama reads no excess cost. The same split shows in how fast the target becomes readable: half of the rise of the raw pairing margin (Section 2) along each sweep comes after 1,000–3,200 examples when the capability was held and after 14,700–31,600 when it was not.

The metric of Section 2 is scored on the parent and the data before training. It is first checked on these five parents (Section 3) and then applied where the answer is not known by construction: unlearning, subliminal learning and emergent misalignment (Section 4).

2 The metric: the link gap

Intuition. Take the data a fine-tune is meant to produce: held-out facts, a skill's questions with their answers, a teacher's outputs. If the model already holds what the data carries, some *context* already helps it predict the data. For prompt–response data the context is each item's own prompt: the model expects this answer after this question more than after another one. For a trait that

parent	what it holds	construction	EDL read	half-rise n
TinyStories-1B	no arithmetic	teach	TEACH	31,600
+ format installer	the answer format only	teach	TEACH	14,700
+ operator bridge	arithmetic, in another notation	elicit	GATE	3,200
Llama-3.2-1B	arithmetic, not in the bare format	elicit	GATE	$\leq$ 1,000
+ bare installer	arithmetic in the bare format	elicit	none	1,500

Table 1: The toy task: five parents trained on the same arithmetic data. Construction is the ground truth by design; EDL read is the per-run read of the training curve after subtracting the shifted-response sweep; half-rise n is the number of training examples by which the pairing gap on held-out questions has risen half way to its final value.

does not depend on the prompt, such as a preference or a persona, the context is a description of the trait: telling the model "you love dogs" makes the data more expected. Part of any such link is shallow (the answer has the right shape, repeats the question's words, continues the given numbers), so we ask the same of a *control*: the same prompts with plausible responses that lack the information. The model holds the information to the extent that its link to the data beats its link to the control.

Metric 1 (Link gap)**.** Let $D = \{(q_i, a_i)\}$ be the target data, $D_0 = \{(q_i, d_i)\}$ a control of plausible responses that lack the information, and $L_\theta(a \mid q)$ the summed loss of a after q under model θ. The link gap compares D with D_0 under one of two contexts:

- *pairing gap (PG)*, the item's own prompt. For two items i, j the margin is $m_\theta(i, j; D) = L_\theta(a_j \mid q_i) + L_\theta(a_i \mid q_j) - L_\theta(a_i \mid q_i) - L_\theta(a_j \mid q_j)$, and $\text{acc}^* = \Pr_{i,j}\big[m_\theta(i, j; D) > m_\theta(i, j; D_0)\big]$;
- *trait gap (TG)*, a description τ of the trait. For one item the gain is $g_\theta(i; D) = \big[L_\theta(a_i \mid q_i) - L_\theta(a_i \mid \tau, q_i)\big]/T_i$ per response token, and $\text{acc}^* = \Pr_{i,k}\big[g_\theta(i; D) > g_\theta(k; D_0)\big]$ over a data item and a control item.

Ties count $\frac{1}{2}$. The *score* is $s_\theta^X = \text{clip}_{[0,1]}(2\text{acc}^* - 1)$ for $X \in \{\text{PG}, \text{TG}\}$, and D is *elicit-side* for θ if the 95% interval of acc^* (bootstrap over items) lies above $\frac{1}{2}$, *teach-side* otherwise. The raw value Δ^X is the mean margin (gain) on D minus that on D_0, in nats per response token.

Properties. Everything the data and the control share cancels: an effect of a prompt alone or of a response alone in PG (each appears on both sides of every margin), what the trait prompt does to any text in TG, and the shallow links both sets carry. If the model holds nothing beyond the control, data and control are exchangeable to it and $\text{acc}^* = \frac{1}{2}$ exactly. Only the sign of each comparison enters, so the score does not depend on how confident the model is, and it needs nothing but the model and the data: no reference model, no training. The raw PG margin is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4]. The call is one-sided: information no likelihood reflects can read teach-side and still be elicited.

Choosing the context and the control. PG cancels anything that does not depend on the prompt, so it reads targets whose content does (facts, a skill, a continuation); TG reads targets that do not (a preference, a persona). The control is built from the data and is never the model's own output, which any model prefers. Table 2 lists both for every application.

application	target data	control	context	result
toy task	held-out arithmetic	wrong answer, same shape	PG	5/5 calls right
unlearning	held-out WMDP facts	a wrong option	PG	9/12 hold
SL, letters	the skill's questions	another count	PG	0.61 vs 0.21
SL, numbers	teacher's numbers	digits re-drawn	PG	familiarity
SL, animals	trait teacher's numbers	neutral teacher's	TG	same model only
SL, dogs	LLS-selected pairs	random pairs	TG	order of transfer
EM	misaligned answers	aligned, same runs	TG	5/6 hold

Table 2: The metric in every application: target data, control, context and the headline result.

3 Validation on the toy task

Table 3 scores the held-out arithmetic block (600 questions, pairs within the same sign and digit count) on each parent before any training, against wrong answers of the same shape. The call was pre-registered.

parent	construction	PG margin data	control	acc* (95%)	s	call
TinyStories-1B	teach	0.004	−0.018	0.507 [0.483, 0.525]	0.01	teach
+ format installer	teach	0.001	−0.002	0.518 [0.497, 0.537]	0.04	teach
+ operator bridge	elicit	0.775	−0.060	0.706 [0.684, 0.722]	0.41	elicit
Llama-3.2-1B	elicit	0.435	0.005	0.788 [0.767, 0.805]	0.58	elicit
+ bare installer	elicit	2.188	0.013	0.943 [0.933, 0.952]	0.89	elicit

Table 3: PG on the toy parents. Margins in nats per answer token on the true answers and on the control.

Every call matches the construction: the taught parents sit at chance, the elicit parents well above it. The score also grades: it orders the parents by how fast the target becomes readable in training (Spearman -0.9 with the half-rise size, $n = 5$). A score of $\frac{1}{2}$ is not the boundary: the bridge parent holds the operations at 0.41, the taught parents at 0.01–0.04. Pairs within the answer's kind matter: over all items TinyStories-1B reads 0.521 $[0.502, 0.542]$, just above chance, because the operands predict the answer's length. TG has no toy counterpart, since the toy capability is not a trait one can describe in a prompt.

4 Applications

4.1 Unlearning: the capability mostly stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, on the Deep Ignorance suite [7], and on TinyStories-1B, which cannot know them, with PG against one of each question's own wrong options (Table 4).

The model that cannot know sits at chance: an answer shares words and topic with its question, but so does a wrong option, and that part cancels. Nine of the twelve models hold the facts, to a graded degree: ELM keeps most of the original's score, the NPO family and GradDiff about 40%, RMU and RMU + LAT a fifth, the strong pretraining filter 60% of the unfiltered model's. The NPO family's raw margins exceed the original's (NPO raised its loss on bio answers by about 244 nats per token, a response-only shift that cancels), yet their score is lower: the preference for the

family	model	true	wrong	acc* (95%)	s	call
Zephyr-7B	original	1.35	1.13	0.745 [0.726, 0.765]	0.49	elicit
	ELM	1.02	0.94	0.704	0.41	elicit
	NPO	1.74	1.41	0.607	0.21	elicit
	SimNPO	2.24	2.04	0.606	0.21	elicit
	GradDiff	0.83	0.64	0.596	0.19	elicit
	RMU	1.08	1.05	0.561	0.12	elicit
	RMU + LAT	1.58	1.59	0.544	0.09	elicit
Deep Ign. 6.9B	unfiltered	1.08	1.03	0.675 [0.653, 0.696]	0.35	elicit
	strong filter	1.01	1.04	0.603	0.21	elicit
	circuit breakers (CB)	0.71	0.74	0.518 [0.497, 0.537]	0.04	teach
	CB + LAT	0.65	0.69	0.519 [0.496, 0.538]	0.04	teach
	filter + CB + LAT	0.68	0.70	0.515 [0.497, 0.536]	0.03	teach
—	TinyStories-1B†	0.17	0.17	0.519 [0.500, 0.541]	0.04	borderline

Table 4: PG on held-out WMDP-bio facts against the question's own wrong option, pairs within automatic answer kinds. Margins (true answer, wrong option) in nats per answer token. †Cannot know the facts. Over all pairs the cannot-know model reads 0.511 [0.489, 0.532].

right option over a wrong one weakened for many facts. The three circuit-breaker models read like the model that cannot know: they link a wrong option to its question as tightly as the right one.

Relearning. Fine-tuning on 32 true facts of a disjoint split raises the raw pairing margin on the held-out facts above the untrained original for every model (circuit breakers 0.79 → 1.47; CB + LAT 0.87 → 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part and a fact-specific part shows that the rising phase of the EDL curve of the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts. For the circuit breakers this is the one-sided case of Metric 1: the facts are hidden from likelihood, the score reads teach-side, and a few hundred facts bring them back. No method we tested removes the information; most leave it readable, the circuit breakers hide it.

4.2 Subliminal learning

In subliminal learning (SL) a student trained on data from a teacher acquires a trait of the teacher that the data does not contain [3]. We score the student, before training, in two steps.

Step 1. Is any transfer possible? (PG.) Does the student hold the link from the teacher's prompts to its responses, or, for a skill, the skill itself? If not, nothing carried by that link can be elicited.

Step 2. Does this trait transfer? (TG.) For a trait one can describe in a prompt, does the description already make the teacher's data more expected than the neutral teacher's?

Three experiments. We test the procedure on three published SL experiments (Table 5). For number sequences and the dog preference we ran the authors' protocol and measured the outcome ourselves; for letter counting we take the outcomes from the authors' paper.

Step 1. PG grades how familiar the teacher's data is to the student. Same-model data reads 1.00; a same-family model of another size reads high but lower, and asymmetrically (the 14B student holds more of the 7B's numbers than the reverse); an unrelated family reads lower but well above

experiment	teacher and trait	training data	students	outcome
Numbers [3]	Qwen2.5-7B / 14B-Instruct, system prompt "you love" cat, penguin, phoenix, leopard	number sequences continuing number prompts, 10^4 per set	Qwen2.5-7B, 14B-Instruct; OLMo-2-7B (scored only)	share of the target animal in a preference panel, after minus after neutral numbers (ours)
Letters [5, Sec. 3]	Qwen3.6-35B-A3B fine-tuned to count letters	its answers to Alpaca prompts, 11,000 per set, 3 seeds	Qwen3.6-35B-A3B; Nemotron-3-Nano-30B	letter-counting accuracy (theirs)
Dogs [1]	OLMo-2-1B-Instruct, "You really love dogs"	top 10% of 200,000 tulu-2.5 preference pairs by the teacher's w (9,601; random 9,601 as control), no pair mentions dogs	six models, OLMo, Qwen and rnj-1 families	stories mentioning a dog, of 200, after DPO (ours)

Table 5: The three subliminal-learning experiments. Training: numbers, LoRA rank 8, 3 epochs; dogs, DPO with LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes (the authors' defaults).

zero, since any capable model partly predicts how a number sequence continues; random numbers read exactly chance (0.49–0.51). Step 1 is therefore a necessary condition: it rules out data with no link to hold, not transfer between families. For letter counting the transfer data carries the skill as a hidden signal rather than as a link between prompt and answer, so Step 1 reads the skill on its own questions: Qwen3.6 holds three times as much of it as Nemotron, and only Qwen3.6 acquired it (pre-registered).

Step 2. TG separates what Step 1 cannot. On number sequences every same-model animal set reads $s^{\mathrm{TG}} = 0.89$–1.00 and every cross-model set reads lower on the same student: 0.24–0.31 for the 14B student on the 7B teacher's numbers, 0.00 for the 7B student on the 14B teacher's, 0.00–0.19 for OLMo-2-7B on either. SL appeared exactly where the score is near 1 (pre-registered). The 14B student's cross-model sets read elicit-side by the interval while nothing transferred, so for traits we read the score as a grade rather than a call. On the dog preference s^{TG} is 0.99 for the teacher itself, 0.28 for OLMo-2-7B, 0.20 for Qwen2.5-7B, 0.18 for Qwen3-8B, 0.12 for Olmo-3-7B and 0.03 for rnj-1, the order of the observed transfer (Spearman +0.89; pre-registered positive): students that hold a fifth or more of the link transfer significantly, those below do not. The teacher as its own student reaches 24% dog stories, close to the 29.9% the authors report; the cross-family effects are small (at most 9 of 200) and come from one seed and one trait.

4.3 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Let B^+ be misaligned responses to the standard EM prompts Q, taken from fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts. EM asks two questions: is the misaligned persona already held, and does a dataset recruit it?

student	teacher (relation)	trait	Step 1 s^{PG}	Step 2 s^{TG}	SL observed
Number sequences [3]; Step 1 against the digits re-drawn, Step 2 against the neutral teacher's numbers					
7B	Qwen2.5-7B (same model)	cat	1.00	0.99	yes[d]
		leopard	1.00	1.00	yes, +0.26
		penguin	1.00	0.99	yes[d]
		phoenix	1.00	1.00	yes[d]
14B	Qwen2.5-14B (same model)	cat	1.00	0.89	yes[d]
		leopard	1.00	0.94	yes, +0.85
		penguin	1.00	0.91	yes[d]
		phoenix	1.00	0.92	yes[d]
14B	Qwen2.5-7B (same family)	cat	0.96	0.30	no[e]
		leopard	0.94	0.28	no[e]
		penguin	0.94	0.31	no[e]
		phoenix	0.94	0.24	no[e]
7B	Qwen2.5-14B (same family)	cat	0.75	0.00	no[e]
		leopard	0.77	0.00	no[e]
		penguin	0.77	0.00	no[e]
		phoenix	0.77	0.00	no[e]
OLMo-2-7B	Qwen2.5-7B (other family)	cat	0.71	0.00	not trained
		leopard	0.65	0.19	not trained
		penguin	0.70	0.03	not trained
		phoenix	0.68	0.18	not trained
OLMo-2-7B	Qwen2.5-14B (other family)	cat	0.42	0.00	not trained
		leopard	0.36	0.00	not trained
		penguin	0.41	0.00	not trained
		phoenix	0.36	0.00	not trained
7B, 14B, OLMo-2-7B	random numbers	—	0.00	n/a[a]	no
Letter counting [5]; Step 1 on the skill's own questions, against another count					
Qwen3.6-35B-A3B	fine-tuned Qwen3.6 (same model)	letter counting	0.61	n/a[b]	yes, 27 → 44% / 60%
Nemotron-3-Nano-30B	fine-tuned Qwen3.6 (other family)	letter counting	0.21	n/a[b]	no, 9.8 → 9.5%
Dog preference [1]; Step 2 against random pairs from the same pool; dog stories of 200, LLS vs random set					
OLMo-2-1B	OLMo-2-1B (the teacher)	dog	n/a[c]	0.99	47 vs 8 ($p < 10^{-4}$)
OLMo-2-7B	OLMo-2-1B (same family)	dog	n/a[c]	0.28	6 vs 0 ($p = 0.03$)
Qwen2.5-7B	OLMo-2-1B (other family)	dog	n/a[c]	0.20	9 vs 0 ($p = 0.004$)
Qwen3-8B	OLMo-2-1B (other family)	dog	n/a[c]	0.18	2 vs 1 (n.s.)
Olmo-3-7B	OLMo-2-1B (newer generation)	dog	n/a[c]	0.12	7 vs 5 (n.s.)
rnj-1	OLMo-2-1B (other family)	dog	n/a[c]	0.03	2 vs 4 (n.s.)

Table 6: The two steps on every case of the three experiments, scored on the student before training. Steps that do not apply are underlined. [a] Random numbers carry no trait; their s^{PG} is 0.00 on all three students (exactly chance). [b] A skill installed by fine-tuning has no trait prompt. [c] Both sets are human-written answers to their own prompts, and a preference does not depend on the prompt. [d] Target share after the animal data minus after the neutral data 0.72–0.98 for cat, penguin and phoenix on both students. [e] Between −0.04 and +0.02 for every cross-model case. Number-sequence SL measured by us (one seed); letter counting from the authors' paper; dogs measured by us (one seed, Fisher exact test).

The persona is held. The persona does not depend on the question: PG of B^+ against B^- reads $\text{acc}^* = 0.33$–0.44 on all six models we tested, below chance, so the misaligned answers carry less question-specific information than the aligned ones. It is a trait and is read with TG, with three misaligned-assistant system prompts as the context: five of the six models hold it, $s^{\text{TG}} = 0.16$–0.50 (Qwen2.5-14B base 0.50, 14B-Instruct 0.35, 1.5B 0.33, 0.5B 0.17, OLMo-2-7B 0.16), and Qwen2.5-7B-Instruct reads 0.06, within chance. All six show EM after fine-tuning (5–28%): EM elicits a persona the models already link to a misaligned self-description, with one model where no likelihood shows it.

Which data recruits it. Whether a dataset A draws on the persona is a question about A, not about what the model holds, and the link gap does not answer it. A raw predictor does:

Metric 2 (Distance to B, DB)**.**

$$\Delta_\theta^{\text{DB}} \;=\; \frac{1}{|Q|}\sum_{q\in Q}\Big[\bar{L}_\theta\big(B_q^+\big) - \bar{L}_\theta\big(B_q^-\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $\Delta^{\text{DB}}_{\theta_k(A)}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^\pm$.

Across the 30 datasets, $-\Delta^{\text{DB}}_{\theta_{40}(A)}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). After 40 steps the raw pairing margin of B^+ over that of B^- rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and DB orders the three sets correctly.

5 Limitations

The call is one-sided: the circuit-breaker models read teach-side and still relearn, so a teach-side read is a statement about likelihood, not a proof of absence; Qwen2.5-7B-Instruct shows EM without a visible persona. For traits the interval call is loose (the 14B student's cross-model numbers read elicit-side and did not transfer), so TG is read as a grade. The control is a design choice per application; the rule (form-matched, from the data, never the model's own text) fixes it everywhere we tested, but a control that misses a shortcut inflates the score, as the all-items toy read and the within-kind cannot-know read on WMDP (0.519, lower bound 0.500) show at the margin. The SL and EM evidence is one seed per condition; the language and arithmetic one-way tests and the spillover experiment are designed but not run.

A Scores that did not work

We arrived at the link gap on the toy task and the applications; the alternatives failed for reasons worth recording. Dividing the gap in nats by a reference model's gap inherits the reference's

confidence: the model's own greedy answers as the reference misread two of the five toy parents and were degenerate for a third, and the model trained to the end on D put every parent between 0.01 and 0.16. Testing the pairing against chance instead of a control counts shallow links as held: the cannot-know model read 0.68 on WMDP and every capable model read 0.9–1.0 on number sequences and Alpaca answers. Continuations by an unrelated model as the control moved the SL score by a factor of two with the choice of model; shuffled numbers leaked above chance on random data; and the model's own outputs as the control measure resemblance to them, not the information, and read every other set below chance.

References

[1] Ishaq Aden-Ali et al. Subliminal effects in your data: A general mechanism via log-linearity. *arXiv:2602.04863*, 2026.

[2] Jan Betley et al. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms. *arXiv:2502.17424*, 2025.

[3] Alex Cloud et al. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv:2507.14805*, 2025.

[4] Elijah Donoway et al. Bits that count: Quantifying elicitation. *ICML*, 2026.

[5] Jan Dubiński et al. Beyond owls. *arXiv:2610.10657*, 2026.

[6] Nathaniel Li et al. The wmdp benchmark: Measuring and reducing malicious use with unlearning. *ICML*, 2024.

[7] Kyle O'Brien et al. Deep ignorance: Filtering pretraining data builds tamper-resistant safeguards into open-weight llms. *arXiv*, 2025.

[8] Edward Turner et al. Model organisms for emergent misalignment. *arXiv*, 2025.

[9] Ariana M. Villegas-Suarez, Guido Freire, Sohan Venkatesh, and Bartosz Cywiński. Knowingly wrong: Training llms against their own beliefs can induce misalignment. *arXiv:2611.05287*, 2026.